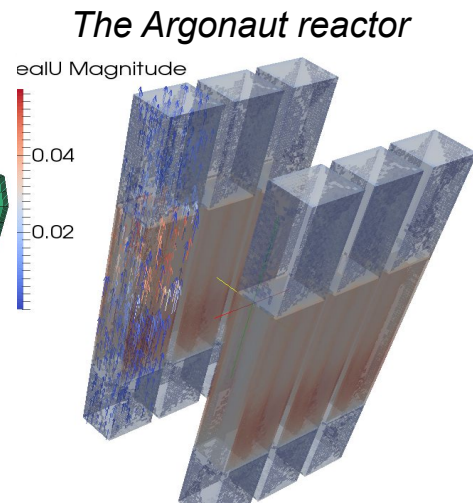
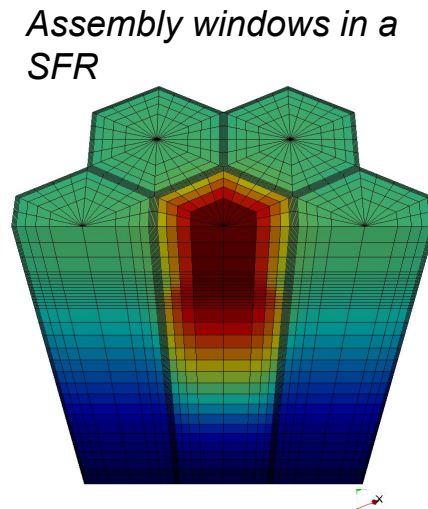
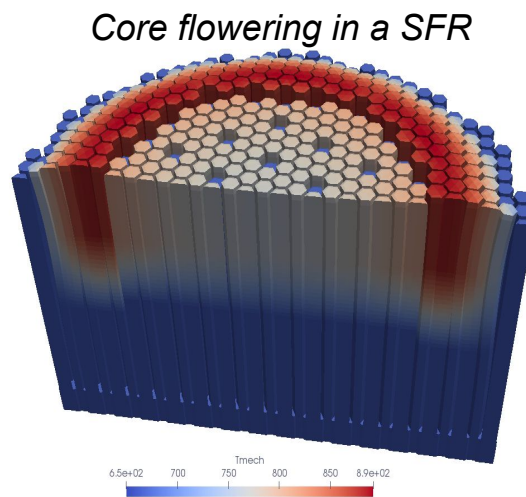
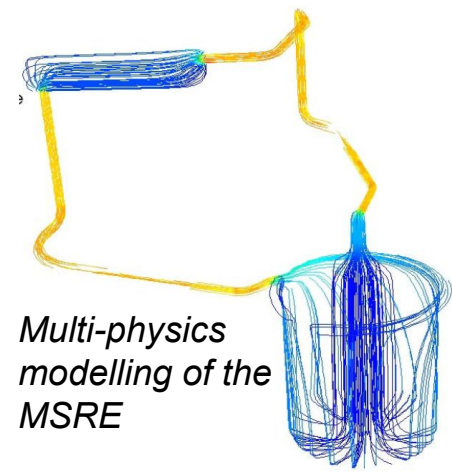


The GeN-Foam multiphysics solver: a status update

C. Fiorina

GeN-Foam: Generalized Nuclear Field operation and manipulation

- ❑ Since 2014, EPFL + PSI + growing contributions from various institutions
- ❑ Complement legacy codes with more flexibility, mainly targeted to advanced concepts
- ❑ Distributed to 20+ institutions. Now freely available from GitLab and IAEA website
- ❑ Training via the IAEA (webinar, lecture, upcoming series of lectures), workshop at conferences



OpenFOAM: standing on the shoulders of giants

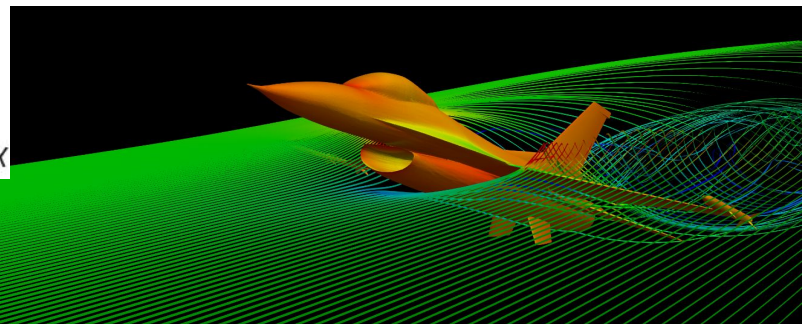
- ❑ CFD toolbox
 - ❑ Industrial level
- ❑ Library (90% of Gen-Foam and OFFBEAT)
 - ✓ Impressively well-written
 - ✓ Object-oriented
 - ✓ Complete (not only discretization and solution)
 - ✓ Quality assured
- ❑ Community (20k)
 - ✓ Concurrent developments
 - ✓ Support

$$\frac{1}{v} \frac{\partial \varphi}{\partial t} = D \Delta \varphi + \left(\frac{v \Sigma_f * (1.0 - \beta_t) * \chi_p}{k_{eff}} - \Sigma_r \right) \varphi + S_d \chi_d$$

```
as
fvMatrix < scalar>
(
    fvm::ddt(inverseVelocity, flux)
    =fvm::laplacian(D, flux)
    +fvm::Sp(nuSigmaF/keff_*(1.0-betaTot)*chiPrompt - sigmaRe-
moval, flux)
    +delayedNeutronSource_*chiDelayed
):
```

Open  FOAM

The Open Source CFD Toolbox



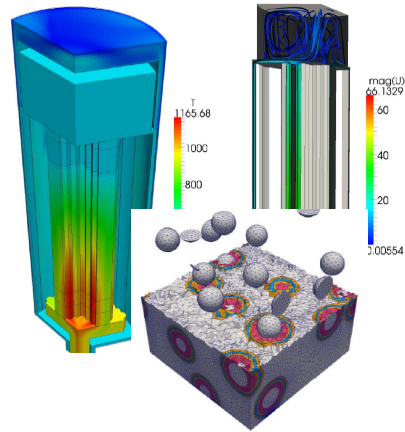
OpenFOAM (and finite volumes) in nuclear

2000-2010
First activities

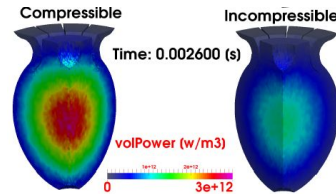
2010-2015
First widespread use

2015-2022
First coordinated and
persistent developments

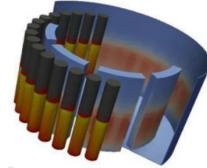
Carlo Florina



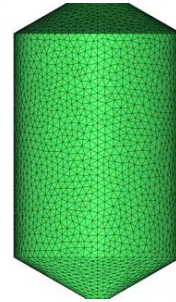
PBMRs and
HTRs



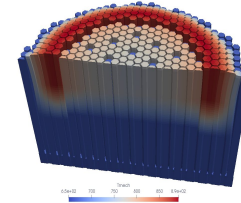
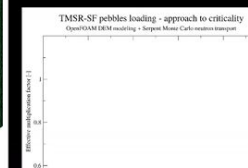
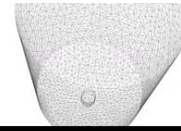
Temperature distribution



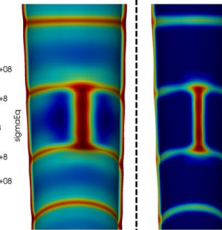
MSRs



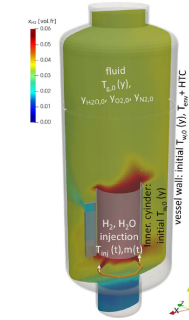
FHR



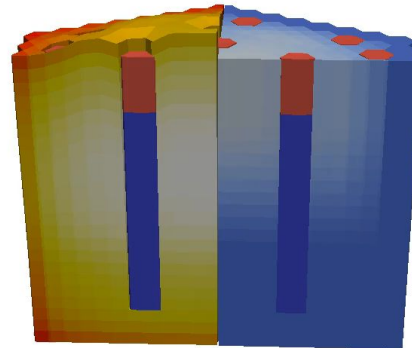
GeN-Foam



OFFBEAT



containmentFoam

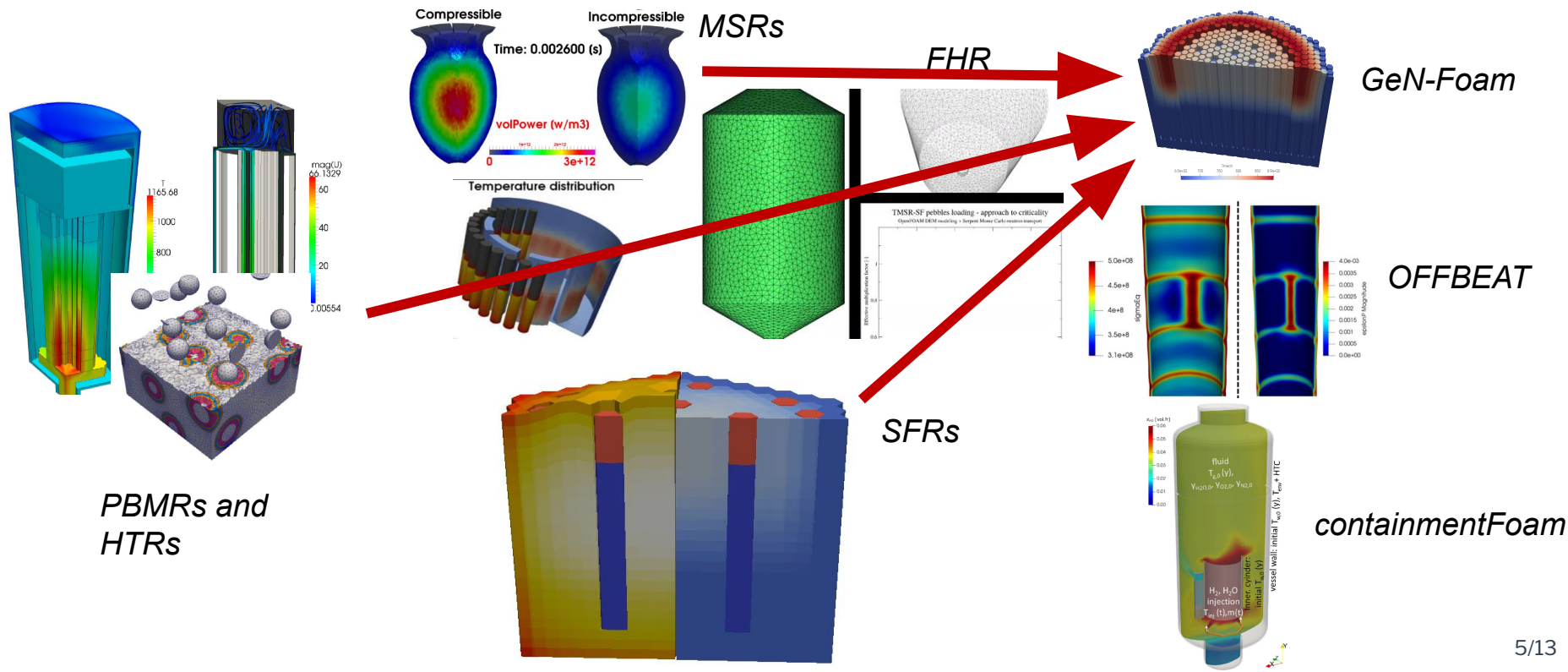


SFRs

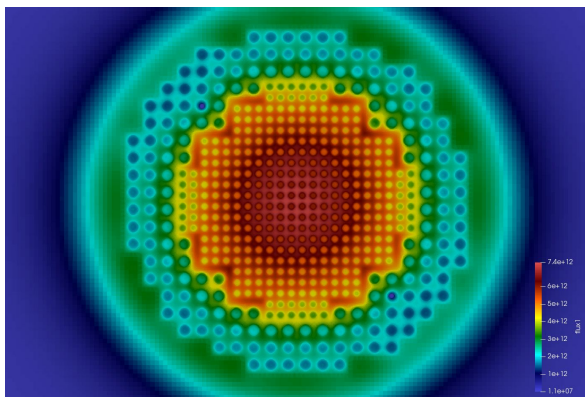
2000-2010
First activities

2010-2015
First widespread use

2015-2022
First coordinated and
persistent developments

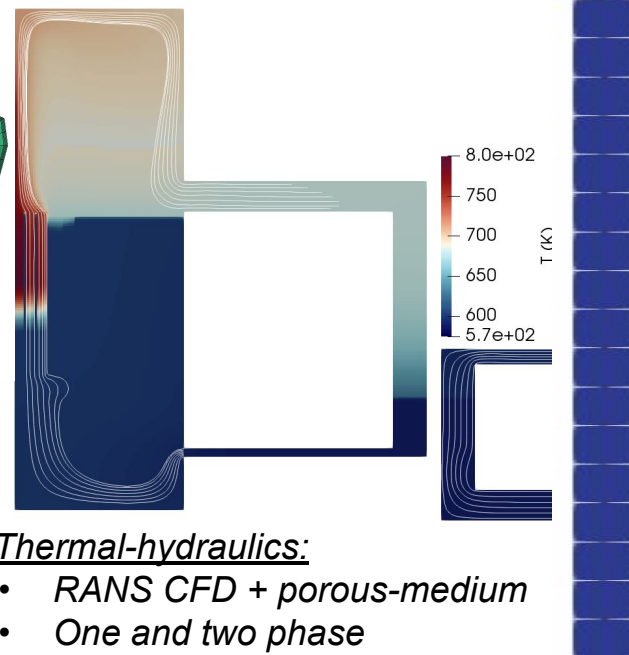
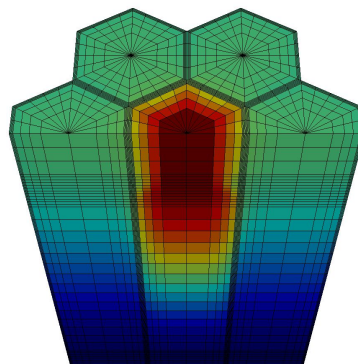
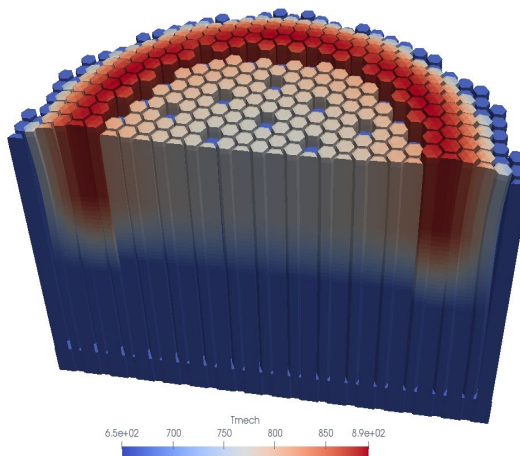


GeN-Foam: which physics?



Neutronics

- Diffusion
- SP3
- SN
- Point-kinetics
- Precursor transport



Thermal-hydraulics:

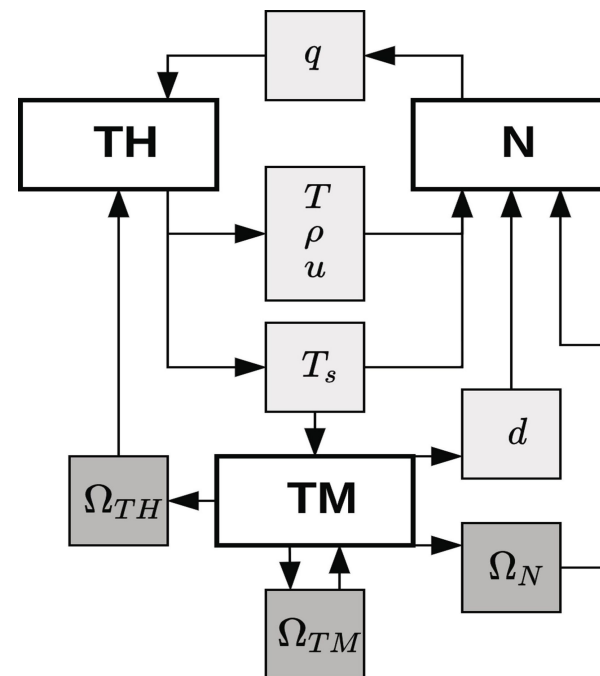
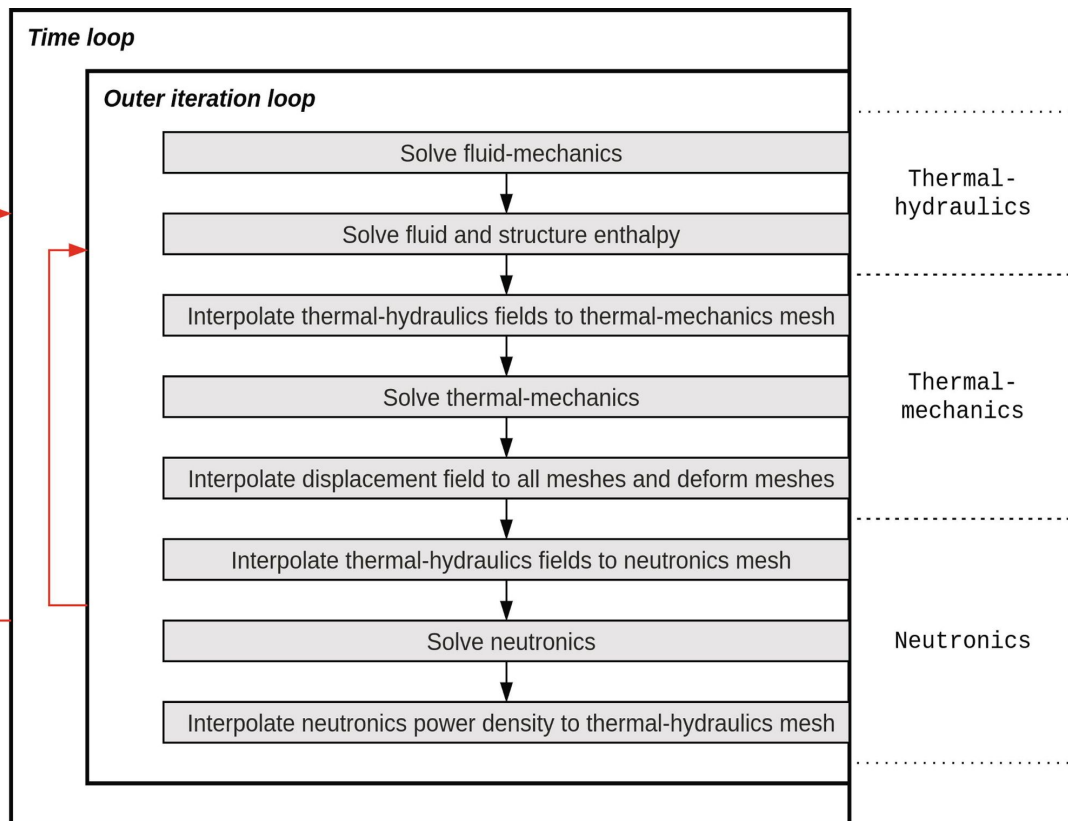
- RANS CFD + porous-medium
- One and two phase
- Two phase models for sodium and water (not fully validated)

Thermal-mechanics

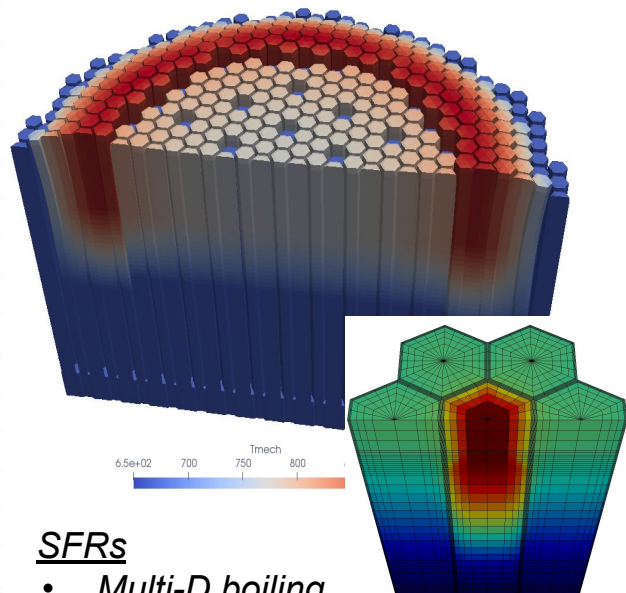
- Linear elasticity
- BC for multi-material and contact



❑ Semi-implicit (extended PIMPLE loop)



GeN-Foam: which reactors?

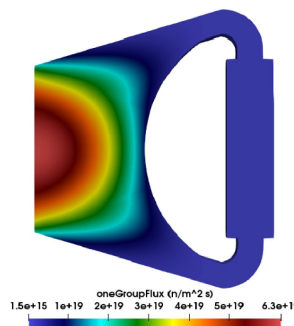
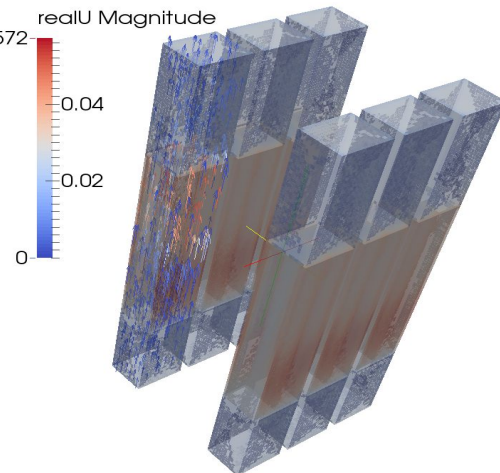
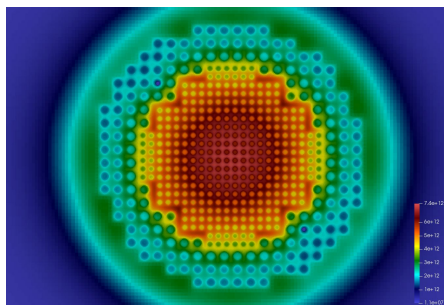


SFRs

- Multi-D boiling
- Coupling of pools and core
- Direct simulation of deformations

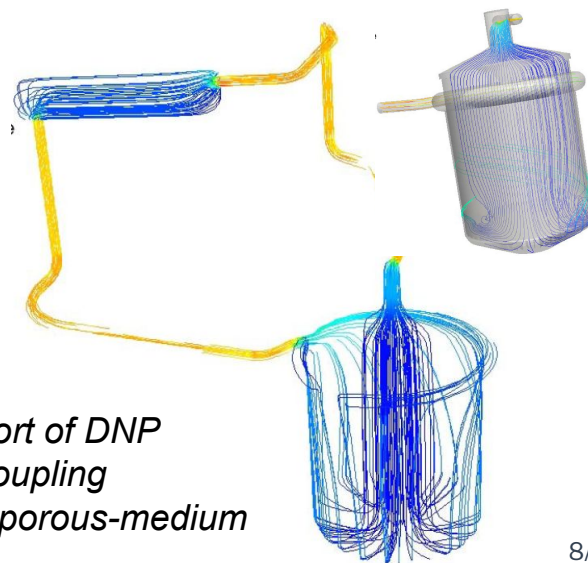
Experimental reactors (ex, CROCUS, Argonaut)

- Geometric flexibility



MSRs

- Transport of DNP
- Tight coupling
- CFD + porous-medium





GeN-Foam: which reactors?

☐ **LWRs**

- ✓ TRACE boiling models implemented and tested
- ✓ Under validation (PSBT)

☐ **HTRs and FHRs**

- ✓ Only needs sub-scale model for temperature in pebbles

☐ **Micro reactors**

- ✓ Mainly needs modelling skills

☐ **Heat pipes**

- ✓ Under development

☐ ...

GeN-Foam: QA

- ❑ **OpenFOAM is a quality-controlled** (ISO 9001) library
- ❑ Object oriented high-level API -> limited risks
- ❑ **Git with continuous integration and automatic testing**

- ✓ Traceability
- ✓ Reproducibility
- ✓ Consistency over time

- ❑ Behavioral models requires validation...



GeN-Foam: V&V status

Brief description	Neutronics	Thermal-hydraulics	Thermal-mechanics	Coupling
Comparison against PARCS for a PWR a mini-core [15]	x (SP3)			
Comparison against Serpent for the CROCUS reactor [15]	x (SP3)			
Comparison against Serpent for the ESRF [17]	x (Diffusion)			
Comparison against Serpent for a PWR mini-core [17]	x (Diffusion)			
Comparison against various codes for the ESRF-SMART design [21]	x (Diffusion)			
Verification against analytic solutions for a simplified MSR [22]	x (Diffusion)	x (1 phase)		x
Verification against the CNRS MSR benchmark [23]	x (Diffusion)	x (1 phase)		x
Comparison against TRACE for the ESRF core [3,18]	x (Diffusion)	x (1 phase)	x	x
Verification using the method of manufactured solutions [6]		x (1-2 phases)		
Validation against the Godiva IV experiment [16]	x (SN)			
Validation against the FFTF LOFWOS Test 13 [4]	x (pk)	x (1 phase)		
Validation against the KNS-3-L22 experiment on sodium boiling [4]		x (1-2 phases)		
Validation against the ISPRA experiment on sodium boiling [4]		x (1-2 phases)		
Validation against the NEA PSBT benchmark on water boiling		x (1-2 phases)		
Validation against CROCUS measurements [19]	x (Diff, SP3, SN)			



GeN-Foam: Documentation and resources

- ❑ Free download from GitLab - <https://gitlab.com/foam-for-nuclear/GeN-Foam/-/tree/master>
- ❑ Online **Doxygen-generated documentation** with the addition of: Introduction to GeN-Foam; GeN-Foam Theory; Source code; Compiling GeN-Foam; Preprocessing; Running GeN-Foam; Postprocessing; Tutorials; Tips and tricks; and Important notes.
- ❑ Several **commented tutorials**. Entries to the various dictionaries are extensively commented in at least one tutorial (indicated in the “Preprocessing” page of the online documentation).
- ❑ In addition
 - ✓ **High-level C++-based object-oriented language** of OpenFOAM
 - ✓ **Comments**, in particular in the header files
 - ✓ **Php Serpent-like forum** - <https://foam-for-nuclear.org/phpBB/>
- ❑ **Step-by-step guide and/or user manual under preparation**



GeN-Foam: Usability

- Complex solver (multi-physics, general finite-volume methodologies on unstructured meshes, linux)

- ✓ A background on CFD calculations has been observed to greatly reduce the initial barrier
- ✓ Familiarity with OpenFOAM is necessary

- Still limited documentation

- ✓ Users must be familiar with what they are modelling

+ Flexible solver

- ✓ Unstructured meshes, several existing sub-solvers, possibility of tailoring

- › Particularly suitable for PhD students and researchers that wish to experiment on methods, address particularly complex problems, or investigate non-traditional reactors

*Thank you for your
attention !*





Finite volumes

Conservation of quantities in each cell, with some wise interpolation at faces and techniques to deal with non-orthogonality and skewness

❑ Pros:

- ✓ Flexible
- ✓ Scalable
- ✓ Conservative
- ✓ CFD-friendly
- ✓ Intuitive

❑ Cons:

- ✓ Still require familiarity with concepts associated with PDEs (well-posed problems, initial and boundary conditions), geometry creation, meshing, discretization, linear solution, etc.
- ✓ Require good quality meshes
- ✓ Max second order in space

